

Towards Personalized Precision Medicine Using DNA-Based Molecular Communication Networks

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ABSTRACT

One of the biggest problems in modern medicine is diagnosing deviations from an individual norm. Most diagnostic methods rely on comparison with a population average. However, this does not account for personal variations. DNA-based nanonetworks offer an alternative approach. Such networks self-assemble from nanoscale components and can compute complex mathematical operations in the process. This paper presents a novel DNA-based nanonetwork architecture that measures several individual health parameters and memorizes them. Based on that, the network determines deviations from that norm and communicates them to an external agent. This works based on already existing technologies, and the ISUTAS software confirms the functionality of the individual components.

CCS CONCEPTS

• **Networks** → **Network architectures**; *Mobile networks*; *Short-range networks*; • **Hardware** → **Biology-related information processing**.

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DNA, DNA-based molecular communication, molecular communication, nanonetworks, nano communication, nano, nanomedicine

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1 INTRODUCTION

Since the dawn of civilization, health has been of major interest. Traditionally, doctors rely on collected experiences with many patients of the past to diagnose a person. Today's treatments are still mostly based on observational studies or based on clinical trials, which have many limitations, such as the historical underrepresentation of women and other groups [1].

However, with the current developments in digitalization and genetic, molecular, and cellular analysis, it is possible to enable diagnostic methods that work on a personal level and are therefore more inclusive and precise. The so-called precision medicine represents a paradigm shift in medicine, from the traditional, very generalized approach to the treatment of a disease to a strategy for the prevention, diagnosis, and therapy of diseases based on a person's unique characteristics [2]. Nanotechnologies are more and more considered to be an important part of precision medicine [6]. Our DNA-based nanonetwork illustrates that nanodevices can help with detecting health parameter anomalies from a *personal norm* instead of a global average, taking precision medicine to the most personalized level possible.

Most research in the area of nanonetwork focuses on top-down approaches where humans take complex machinery to actively manufacture devices. This approach has limitations at the nanoscale, and we focus on an alternative paradigm that has been tested by nature itself: self-assembly.

We consider this to be the most promising approach on the nanoscale because of its extremely small size alone. Working at the molecular level makes top-down design and production of man-made objects extremely difficult. However, if it were possible to simply throw the building blocks of the objects together and wait for the desired objects to form, this problem would be elegantly solved.

Fortunately, nature has already invented such a basic mechanism: Crystals. Crystals can be understood as a lattice structure of simple building blocks. Tiny and often simply-structured building blocks join together to form larger and more complex structures. Unfortunately, on the other hand, most crystal formation processes cannot be properly controlled. Exceptions are certain building blocks of *deoxyribonucleic acid* (DNA). Conditions for the growth process of DNA crystals can be encoded by the correct choice of constituent bases. It is even possible to make the necessary DNA strands in the laboratory. Consequently, the generation of DNA building blocks that behave like puzzle pieces is feasible [9, 11]. These DNA puzzle pieces are called tiles and can combine in predetermined ways to create almost any nanoscale structure [5, 11]. In [4] we have shown the assembly of all components for fully functional DNA-based networks from a single building block: DNA tiles. This guarantees the compatibility of construction, computation, and communication for complex nanonetworks, which is difficult with other technologies right now.

We further explore the idea of using those networks to solve outstanding problems in the area of medicine in this paper. This is feasible due to the fact that DNA-based nanonetworks are Turing-complete and can compute any mathematical operation a macroscopic computer can perform. While they are not programmable like a classical Von Neumann computer, it is still possible to design a new network for each problem. DNA-based nanonetworks are comparable with hardwired electronic circuits.

We demonstrate a DNA-based nanonetwork that measures a number of health parameters of a patient and stores them as tiles in the communication channel. Based on the measured data, the network can enter the second mode of operation, where it detects anomalies concerning the previously determined values. It can either immediately react to the measurements by releasing a previously-stored medical payload or by communicating

the results to an external device or person. We thus solve several problems at once and demonstrate that DNA-based nanonetworks are more powerful and versatile than classical medical tests in almost every way. They can compute, have memory, and can act completely autonomously if necessary.

Based on those capabilities, it is possible to design medication that only has an effect if the recipient is really sick. Another option is a class of medication that only applies if other health parameters are within a tolerable range or medication that only works if no harmful prior conditions apply. Yet another use case is the design of a diabetes medication that is only triggered once blood sugar reaches critical levels.

The remainder of this paper is organized as follows: Section 2 explains the necessary mathematical basics in brief. Without that knowledge, understanding the presented network designs is impossible. Section 3 explains the novel concept for personalized precision medical applications and its network architecture in detail. We highlight and solve many challenges towards the conceptualization of such systems. Section 4 verifies the correctness of the presented nanonetworks using the ISUTAS simulation software. Section 5 concludes the paper and explains several (no longer so) futuristic scenarios and ideas that can be realized based on the presented network architecture.

2 RELATED WORK

It is possible to describe self-assembly systems as crystal-like growth processes. Thus, they form a solid base for analyzing and simulating such processes, and they can help create wet-lab implementations. This section teaches a basic understanding of the mathematical foundations of self-assembly systems.

2.1 Introduction to Self-Assembly Systems

This section presents brief explanations as well as examples of systems for self-organization based on the work of Eric Winfree [11]. The notation used below is based on [3] and [4]. Detailed definitions can be found in [4].

The basic components of systems for self-organization are called *tiles*. Tiles are generally defined as mathematical objects in $\mathbb{Z}^n$. The most important applications are in $\mathbb{Z}^2$ and $\mathbb{Z}^3$. For simplicity, we assume $\mathbb{Z}^2$ unless otherwise stated. A tile t is a square in $\mathbb{Z}^2$ with glues on each side. Figure 1 (a) and (b) show examples of two-dimensional tiles. Figure 1 (a) shows a representation that focuses on the mathematical model. Figure 1 (b) highlights the biological building blocks and interactions in a DNA-based implementation.

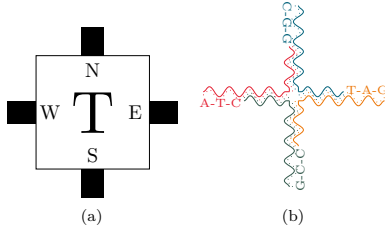


Figure 1: Examples of a tiletype in 2D – mathematically and biologically modeled. The mathematically modeled glues are represented by black boxes, their color by a label. The biologically modeled glues are represented by base sequences.

An important property of tiles is their ability to bond with other tiles. For this purpose, the tiles are equipped with so-called glues. Each side of a tile can have a number of glues. These are represented by the corresponding number of black boxes, as in Figure 1 (a). The number of glues is called *strength*. Each glue has a property called *color*. The colors of the glues are distinguished by different labels on each side. If a glue is not explicitly shown or mentioned, a glue with strength 0 and no color is assumed.

In DNA-based tile systems, glues are realized by open DNA strands with self-assembling base sequences [8]. The DNA can be used for both color and strength. Color is encoded by the different bases and strength by the length of an open DNA strand.

In addition, the tiles have a marker in the middle, which is represented by a character. This allows a tile to be associated with a truth value to interpret the result of calculations. The markers serve as a feature to distinguish DNA tiles from each other instead of laboriously comparing the glues when analyzing simulation results.

Each tile occupies exactly one *position* in the discrete two-dimensional space $\mathbb{Z}^2$. Tiles are *neighbors* if they occupy adjacent positions. A tile can have no neighbor or exactly one neighbor on each side. Tiles t and t' interact with each other only if they are neighbors. The interaction rules are determined by the glues as follows: Adjacent tiles with matching glues can bind to each other. Two glues match if they have the same color and thickness. Multiple tiles that form a contiguous bond are called *assembly*.

The bond is subject to a *temperature* τ . The temperature describes the minimum bond strength required to form a stable bond. If the temperature is 2, two matching black boxes of matching color are required for a tile to form a *correct* bond with an assembly. A tile t that has

a glue of strength 0 or no glue on its side cannot interact with adjacent tiles of an assembly. The temperature τ models the physical temperature. Molecules move faster when the system temperature is raised. The extra energy makes it more likely that bonds will be broken.

Tiles are assumed to be infinitely available. A *tiletype* groups together all tiles that are identical. Tiles that have the same glues and bond strengths and the same markings are of the same *tiletype*.

3 PERSONALIZED NANONETWORKS

While humans overall are similar, the normal ranges of various health parameters differ from person to person. As an example, the blood pressure of humans who lift heavy weights is naturally elevated. However, that is usually not critical and just a bodily adaptation to weight training. For other people, the same value could be dangerously high, and blood pressure is just one of many examples. Generally, many health parameters follow a Gaussian distribution.

In classical medicine, medication is usually determined by comparison to a deviation from the population average, and the treatment follows a one-drug-fits-all model. While this can be a useful tool for the population average, it fails to account for individual variation. Especially people who deviate from the average can't be reliably diagnosed. Remarkably, this deviation from the average basically affects every woman, for which the hormonal cycles can influence various bodily parameters. Despite being the more complex individuals, women are excluded from most clinical trials, and the results obtained solely by examining men are extrapolated to women [1].

To solve this problem of generalization, we conceptualize a nanonetwork that first measures normal values for an individual and later detects anomalies in comparison. Thus, we present a functional nanonetwork design to solve this general problem while also allowing for immediate treatments. We conceptualize this network as a powerful medical test outside of human bodies but envision it to work inside living beings as well.

3.1 Phase 1

We divide the task of the nanonetwork into two main phases: first, measuring, and later monitoring and treatment. Figure 2 divides the first of the two phases into five subparts. The entire process starts once all tiles and the seed-tile σ are present in the medium. Once started, nanosensor¹–nanosensor ^{n} detect different health parameters and release specific tiles upon detection. This continues for a time t where message molecules of arbitrary length can form in part 3. Once time t elapsed,

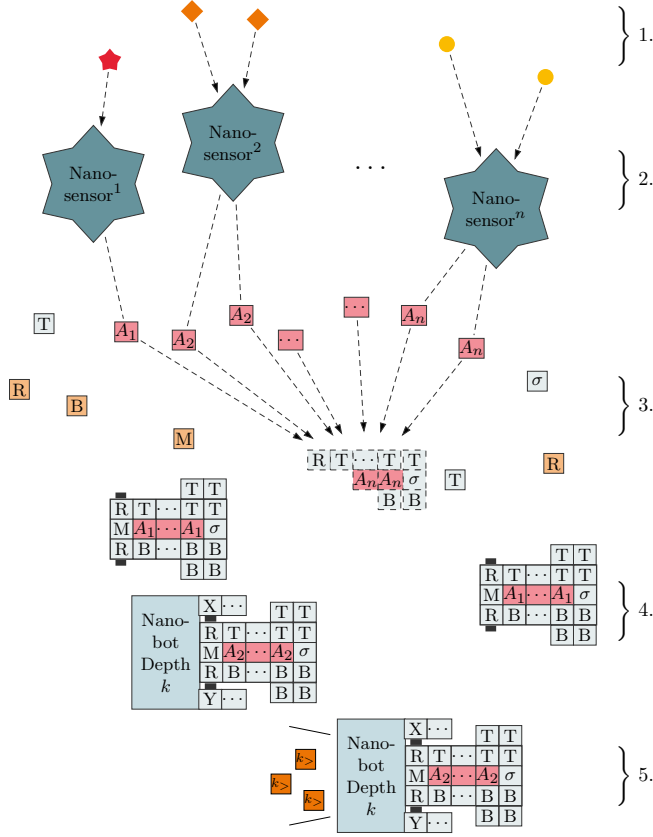


Figure 2: In phase 1, the purpose of the nanonetwork is to determine the normal range of body parameters 1– n . Nano-sensor₁ counts the frequency of health parameters and releases tiles. In the medium, messages of arbitrary length form that are only stopped once a receptor tile R has been added externally. Now, the length of the message represents a unary number that stands for the normal range of parameter i over time t . The messages can now fully assemble and only bind to DNA-boxes of depth k or lower. Upon binding, the content of the boxes is released and now represents the threshold information as a single tile.

the orange tiles R, B, and M are introduced and stop the assembly process by adding a receptor to the message molecule that prevents further assembly. The thus assembled message represents the concentration k of a measured parameter in unary coding.

The fully assembled messages can then bind to fitting boxes of maximum depth k like a key. Each of the opened boxes releases new seed tiles k_+ for phase 2. The boxes also release tiles that bind to all seed tiles $k - i_+$ and thus ensure the presence of only a single seed tile that

represents a normal health parameter. Thus, the measured concentration is converted from unary coding to a single tile. It is also possible to release signal repeater tiles from the same boxes to amplify a weak signal. After this step, the network automatically enters phase 2.

3.2 Phase 2

Phase 2 is depicted in Figure 3. The initially released seed-tiles k_+ begin forming message frameworks of length k plus a predefined threshold j . For the same time interval t , the messages are now allowed to fill the assembled frame before the medical test expires. If the messages fully assemble in the given time, this signifies the reaching of a critical, predefined threshold. The finalized messages can then bind to a different type of nanobot that either immediately reacts by releasing medication where it is needed or by communicating the measurements to the outside.

The functionality of the concentration counter can be seen in Figure 4 (a). A frame of height $k + j$ grows once seed tile k_+ starts the process. The tiles A_i have to fill the frame completely for the left part of the receptor R to form.

In the visionary scenario inside of the human body, the second phase has to be repeated continuously. It is worth mentioning that only the exceeding of a threshold may be detected. In DNA-based nanonetworks, we cannot tell apart if molecules will never finish or are *not yet* finished. Thus, we need a different approach to detect if too little of a health parameter is detected. One possibility is the engineering of a molecule with a finished receptor that gets blocked once enough tiles are present.

4 EVALUATION

We use the ISUTAS software from Patitz [7] to evaluate the proposed assemblies. The molecules shown in Section 3 are instantiated in the corresponding modules for kTAM and HAM.

We first employ the kTAM model to simulate assembly processes under realistic conditions. The kTAM sensibly predicts wet-lab experiments, as shown in [10]. Currently, DNA-based nanonetworks require laboratory conditions for the assembly process as precise modulation of the environmental temperature is required. However, it is possible to use DNA-based nanonetworks in controlled environments like a petri-dish to analyze, for example, tissue samples.

Afterward, we analyze the tilesets with the 2HAM module of the ISUTAS software. It tests all possible interactions between tiles and intermediate assemblies. We use the 2HAM to confirm the binding between nanobot

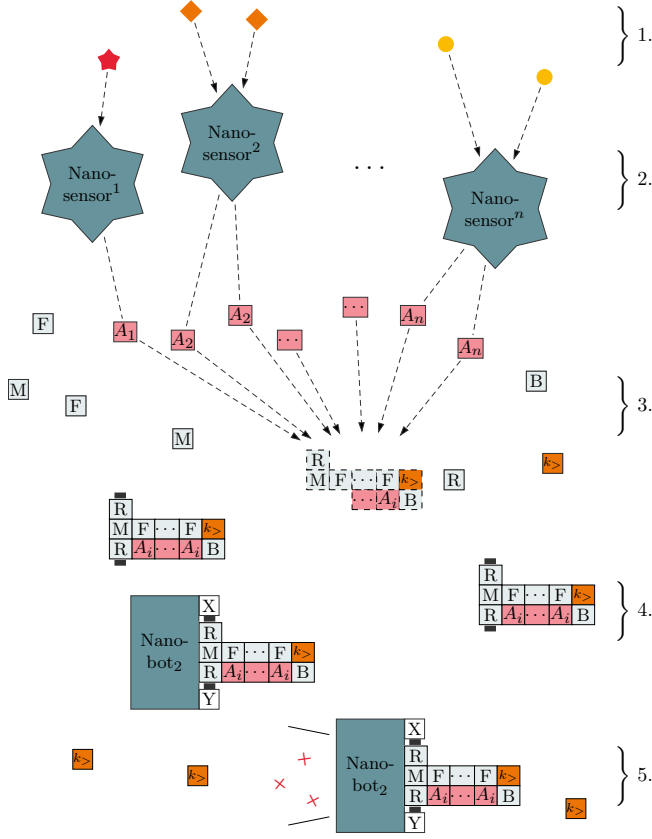


Figure 3: In phase 2, the purpose of the nanonetwork is to measure deviations from the previously determined normal values. The message molecules in the middle can only fully assemble if a threshold has been violated by a predefined value. Upon such an event, an appropriate Nanobot is opened and the measurement communicated to an external party. Alternatively, the nanobots directly fix the anomaly.

receptors and message molecules and to investigate possible error sources of intermediate assemblies.

4.1 Realistic Simulation in the kTAM

We use kTAM parameters $G_{se} = 10.4$ and $G_{mc} = 17.0$ for the simulation to match the experiments from [10]. G_{se} is the costs to break a binding and G_{mc} stands for the monomer concentration. $\frac{G_{mc}}{G_{se}}$ roughly models the temperature parameter τ from the aTAM. We simulate “difficult” self-assembly conditions, as $G_{se} = 10.4$ has led to numerous errors in the simulation in [10]. $G_{mc} = 17.0$ model a tile concentration of $0.8 \mu\text{M}$. This is a comparable concentration to the sodium in human

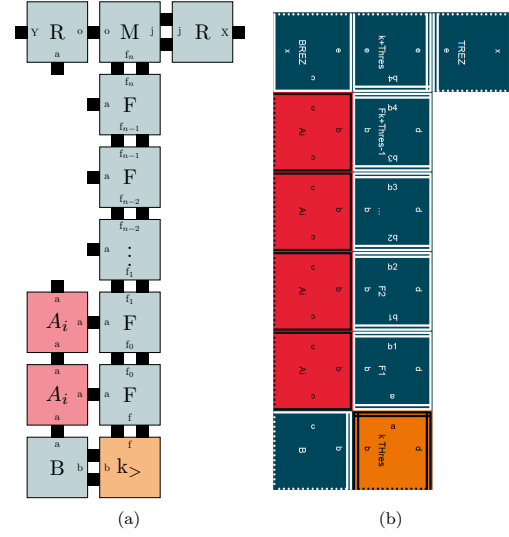


Figure 4: (a) Assembly of a message molecule for the counting problem with a ligand. The blue frame structure demands up to $k + \text{thres}$ fitting tiles to bind before the ligand can be completed. (b) The result of the simulation of the same message molecule.

blood [4]. A value $G_{se} = 10.4$ further corresponds a temperature of 32.7°C . Lower values stand for higher temperatures and make binding reactions less likely. Each tiletype occurs infinitely often in the medium with the same probability and distribution. These are reasonable assumptions for in vitro conditions but do not describe environments in the human body well. [4] offers some insights on already conducted wet-lab experiments.

The syntax of the simulator should be understood as follows: The text in the middle is the marker and serves the purpose of distinguishing tiles. A dotted line at the side means np glue. One line with a label next to it is a glue of strength one, and two lines with a label a glue of strength two.

Figure 4 (b) shows a fully assembled message molecule that counts the concentration of a health parameter in unary coding plus a threshold l . We chose a value $k = 3$ and an additional threshold of $l = 2$. Tile $\dots$ could represent an arbitrary value k . The message molecule assembles without any growths or facet errors [8].

4.2 Analysis in the 2HAM

In this section, the interaction of intermediate assemblies and receptors/message molecules is examined in the 2HAM. The simulations in the 2HAM show that the assembly process reaches a fixpoint upon which no additional intermediate assemblies can form. Furthermore,

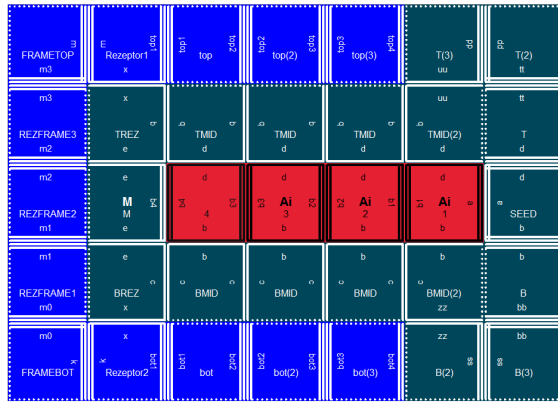


Figure 5: Approximation for the message molecules that counts tiles in unary coding. Once finished, it is correctly bound to a fitting box, modeled as a receptor, of appropriate depth k . Any shorter message molecule can not reach the receptor inside the tube to trigger the release of threshold tiles.

the proposed message molecule from phase 1 binds to a nanobot without problems. It is not possible for the same molecule to bind to a deeper nanobot as the thick end of the message molecule blocks in from reaching the glues inside the receptor.

5 CONCLUSION

This section concludes the paper by summarizing the most important points and results. Furthermore, we outline possible (visionary) future applications of DNA-based nanonetworks.

5.1 Summary

The proposed nanonetwork can measure a concentration value and persistently store the measured value as a tile. The network learns a “normal range” for a number of health parameters and can directly react upon exceeding a predefined range in phase 2. This approach is superior to regular medical tests, as both memory and computational power can be utilized. The test is of a very general nature and can be adapted to various parameters or disease markers. This type of network thus theoretically allows us to solve yet unsolved problems.

5.2 Future Work

There are many possibilities to build on this important stepstone towards personalized precision medicine. In the future, it might be possible to directly produce mRNA sequences upon the detection of a problem to create a vaccine and assist the human immune system. It might

even be possible to send the blueprint for the vaccine to a nanonetwork inside the body of other people to immunize them as well. Based on the data collected, creating a database of “normal ranges” for various people can be an important building block to improve future diagnostic capabilities. And lastly, the proposed network might be enhanced by a clock system that periodically triggers phase 2 to make it work even in a living organism.

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